

EpiCite: A Two-Stage Epistemic Sentence Classifier and Citation-Need Scorer with Triangulated Feature Attribution

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Abstract—We present EpiCite, a two-stage NLP pipeline that first classifies sentences by their epistemic role (Claim, Evidence, Opinion, Background) and then scores their citation need. Stage 1 is a DistilBERT classifier trained with focal loss ($\gamma=2.0$) and a class-balanced sampler on a merged IBM+UKP corpus of 35,306 sentences, reaching a macro-F1 of 0.763 on the held-out test split. Stage 2 is a Late-Fusion citation scorer that concatenates DistilBERT’s [CLS] embedding (768-dim) with a 29-dim engineered feature vector and Stage 1’s five epistemic probabilities, trained on 150k balanced WikiSQE sentences with three-phase gradual unfreezing, reaching ROC-AUC 0.727 (F1 0.655). A four-axis evaluation suite (SHAP, permutation importance, mutual information; integrated gradients; class-conditional SHAP; reliability diagrams; an ablation; a Qwen-0.5B benchmark) yields three honest findings. (1) A triangulated Borda-rank feature analysis ranks `sentence_length` first by a wide margin and places every epistemic probability at rank 9 or below, refuting our hypothesis that epistemic type dominates surface features. (2) An ablation removing the Stage 1 signal slightly improves ROC-AUC (+0.010), so the explicit two-stage design contributes interpretability rather than accuracy. (3) Despite this, EpiCite outperforms a $3.7\times$ larger Qwen-0.5B-Instruct LLM by +0.19 AUC at $\sim 11\times$ lower latency, with errors that are near-independent ($\kappa = -0.005$), suggesting ensemble potential. We release the pipeline alongside an MCP server demo for sentence- and paragraph-level audit.

Index Terms—citation-need prediction, epistemic classification, late fusion, feature attribution, SHAP, integrated gradients, model interpretability, NLP

I. INTRODUCTION

Verifiability is the cornerstone norm of Wikipedia and an increasingly important property of any text consumed by automated pipelines. Detecting sentences that lack but require a citation is the core task formalised by the LSTM baseline of Redi et al. [1], who treat it as binary classification over surface features such as sentence length and position. While effective on Wikipedia, such models cannot explain *why* a sentence was flagged: they fail to distinguish an unverified factual Claim from a subjective Opinion or an encyclopedic Background statement [2]. This opacity blocks reliable transfer to other writing genres (news, student essays) where the norms of verifiability differ from Wikipedia’s [3], [4].

We propose **EpiCite**, a two-stage pipeline that explicitly models epistemic sentence type as a precursor to citation scoring (Fig. 1). Stage 1 is a focal-loss DistilBERT that labels each sentence as Claim, Evidence, Opinion, or Background, with a class-balanced sampler addressing the $65\times$ imbalance between Claim and Evidence in the combined IBM+UKP corpus. Stage 2 is a Late-Fusion scorer that fuses DistilBERT’s [CLS] embedding with a 29-dim engineered feature vector and the five epistemic probabilities, training in three gradual unfreezing phases with cosine-annealed learning rates.

The motivation for the two-stage design is twofold: Bhagavathula et al. [5] showed that decoupling candidate selection from discriminative scoring improves citation precision, and Cohan et al. [2] reported a 13.3% absolute F1 gain on ACL-ARC from adding an intermediate structural-role task. We validate, refine, and where necessary refute these intuitions.

a) Contributions:

- 1) A class-balanced focal-loss DistilBERT for four-class epistemic classification that reaches macro-F1 0.763 with a $65\times$ imbalance.
- 2) A 29-feature engineered vector (lexical, syntactic, entity-rate, numerical, and verb families) and a Late-Fusion architecture that concatenates these with [CLS] for citation scoring.
- 3) A **triangulated** feature ranking (SHAP, permutation importance, mutual information; aggregated with Borda count) and a **class-conditional** SHAP analysis that reveal which features matter *per epistemic class*.
- 4) An **ablation** that quantifies the contribution of the Stage 1 signal to the final scorer, alongside a head-to-head comparison with Qwen-0.5B-Instruct on classification quality, latency, and interpretability.
- 5) A reproducible MCP server interface exposing `analyze_sentence`, `analyze_paragraph`, and `explain_prediction` for downstream audit tools.

b) *Hypotheses*: **H1**. Epistemic probabilities (especially $P(\text{Claim})$) outrank surface features such as sentence length as predictors of citation need. **H2**. The two-stage Late-Fusion design offers higher *interpretability* than an end-to-end DistilBERT classifier through feature-level SHAP and token-

level integrated gradients. **H3.** Removing the Stage 1 signal degrades scorer performance (ablation justification).

c) *Research questions:* **RQ1.** Does Late-Fusion with epistemic features improve over a no-epi variant? **RQ2.** Which engineered features dominate, agreed by multiple independent attribution methods? **RQ3.** How does EpiCite compare to a generative LLM (Qwen-0.5B) on citation-need prediction across accuracy, latency, and interpretability?

II. RELATED WORK

a) *Wikipedia citation need:* Redi et al. [1] predict the `{{citation needed}}` tag with an LSTM over surface features, the canonical task we extend. The recent WikiSQE benchmark [7] provides the large-scale labelled split we use for Stage 2.

b) *Structural role and citation intent:* Cohan et al. [2] demonstrate that an auxiliary structural-role task (background / method / result) lifts citation-intent classification by 13.3% absolute F1, providing the strongest prior evidence that an epistemic intermediate layer can help. Zhang et al. [3] make a parallel argument that conventional models underuse structural and lexical context.

c) *Two-stage citation pipelines:* Bhagavatula et al. [5] decouple candidate generation from discriminative scoring in citation recommendation. We adopt their staged philosophy but apply it to the upstream *should-this-be-cited?* question rather than to recommending *which* reference to cite.

d) *Linguistic features of factuality:* Soni et al. [6] survey hedging, modality, and other factuality markers. We operationalise this literature as 22 of the 29 engineered features used by Stage 2.

e) *Attribution methods:* We adopt SHAP [10] for global feature ranking and Integrated Gradients [11] for token-level attribution; LIME [12] is used as a sanity check. To our knowledge, EpiCite is the first citation-need work to triangulate attributions across SHAP, permutation importance, and mutual information under a single Borda aggregation.

III. DATA

a) *Stage 1 — epistemic labels:* We merge the IBM Claim Detection corpus [8], IBM Evidence, IBM Context Dependent Claims, and the UKP Sentential Argument Mining corpus [4], remapped into the four-class Claim/Evidence/Opinion/Background taxonomy. The *combined* pool (35,306 sentences) is summarised in Table I; note the substantial uplift in Evidence (from 204 to 3,949 sentences) brought by adding IBM Evidence. We additionally retain a separate IBM-only held-out test set (878 sentences) to measure source-distribution generalisation.

b) *Stage 2 — citation labels:* WikiSQE [7], `citation_needed` configuration, with the provided train/val/test splits. We subsample 150k balanced sentences (75k positive, 75k negative) for training to fit a single-GPU budget; the official test split (991 sentences) is left untouched for all reported metrics. The full WikiSQE corpus we processed contains 151,981 sentences across all splits.

TABLE I
STAGE 1 TRAINING POOL: ORIGINAL (UKP/IBM-CDC) VS. COMBINED (IBM-AUGMENTED). CLASS-BALANCED SAMPLING IS APPLIED AT TRAINING TIME.

Class	Original	Combined	Δ
Claim	13,225 (48.3%)	17,388 (49.2%)	+4,163
Evidence	204 (0.7%)	3,949 (11.2%)	+3,745
Opinion	2,433 (8.9%)	2,433 (6.9%)	0
Background	11,536 (42.1%)	11,536 (32.7%)	0
Total	27,398	35,306	+7,908

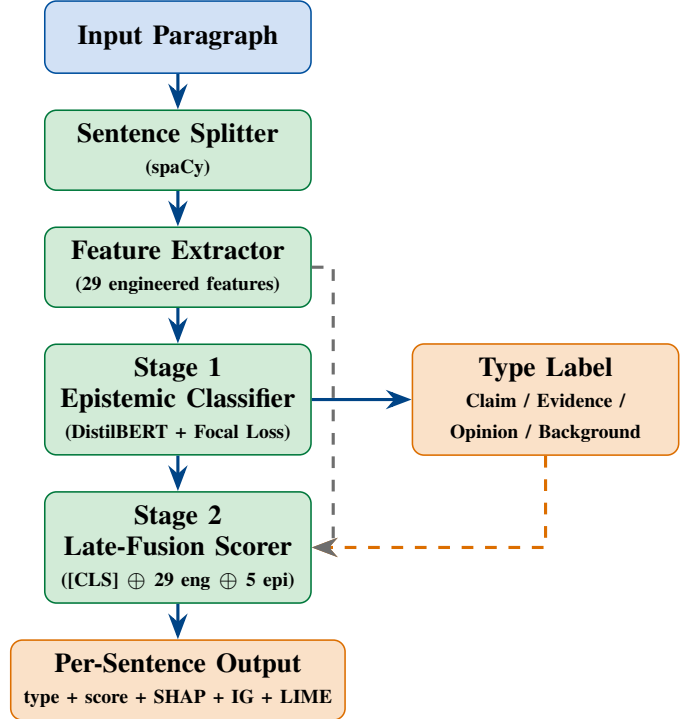


Fig. 1. The EpiCite two-stage pipeline. Dashed orange: Stage 1 epistemic probabilities are passed to Stage 2. Dashed gray: engineered linguistic features are also passed to Stage 2 alongside the `[CLS]` embedding.

IV. METHODOLOGY

A. Engineered Feature Vector (29-dim)

We use spaCy and NLTK to extract 29 interpretable features per sentence, grouped into six families (Table II). The hedging, boosting, and quantifier lexicons follow the factuality survey of Soni et al. [6]; the entity-rate, numerical, and verb-class families are added in this work to enrich the 12-feature baseline used in our preliminary version.

B. Stage 1 — Focal-Loss Epistemic Classifier

We fine-tune DistilBERT (distilbert-base-uncased; 66M params) with multi-class focal loss [9]:

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_c (1 - p_t)^\gamma \log p_t, \quad \gamma = 2.0,$$

where α_c is an inverse-frequency per-class weight ($\alpha_{\text{Claim}}=0.571$, $\alpha_{\text{Evidence}}=1.199$, $\alpha_{\text{Opinion}}=1.528$,

TABLE II

ENGINEERED FEATURE FAMILIES (29 DIMS). FIVE STAGE 1 EPISTEMIC PROBABILITIES ARE CONCATENATED SEPARATELY FOR THE FULL STAGE 2 MODEL (34 DIMS TOTAL).

Family	Features	N
Hedging	hedge verbs, hedge auxiliaries, hedge adverbs, vague quantifiers, boosters, modals, negations	7
Subjectivity	subjective pronouns, reporting verbs, causal markers, universal quantifiers, superlatives	5
Entity rates	PERSON, ORG, GPE, DATE, total sentence %, dollar, year, digit, inline-citation rates, proper noun count, raw number count	5
Numerical		7
Syntactic	passive voice flag, dependency-tree depth, clause count, noun/verb ratio	4
Length	sentence length (tokens)	1
Total engineered		29
Epistemic	Stage 1 class probabilities + max confidence	5
Stage 2 input dim [CLS]	(excl.)	34

$\alpha_{\text{Background}}=0.702$). Focal loss down-weights confident easy examples, letting the rare-class gradient dominate optimisation. We additionally apply a `WeightedRandomSampler` so each batch sees a class-balanced distribution. Training uses AdamW ($\text{lr} = 2 \times 10^{-5}$), cosine annealing with 10% warmup, batch size 32, max 8 epochs, early stopping with patience 2 on validation macro-F1. The model is warm-started from a class-weighted DistilBERT checkpoint.

C. Stage 2 — Late-Fusion Citation Scorer

The scorer concatenates DistilBERT’s pooled [CLS] embedding (768-dim) with the 34-dim engineered+epistemic vector and feeds the 802-dim result through a 3-layer MLP with LayerNorm and dropout (0.3, 0.2), ending in sigmoid:

$$\text{Linear}(802 \rightarrow 256) \rightarrow \text{Linear}(256 \rightarrow 64) \rightarrow \text{Linear}(64 \rightarrow 1).$$

LayerNorm (rather than BatchNorm) avoids train/eval discrepancies when fine-tuning the transformer jointly with the head.

a) Three-phase gradual unfreezing: A randomly initialised head co-trained with a pretrained transformer yields noisy gradients that corrupt transformer weights. We therefore train in three phases with cosine annealing (Table III); patience-2 early stopping is applied within each phase.

D. Explainability — Triangulated Attribution

A single attribution method can mislead. We therefore evaluate feature importance with three independent estimators on the 991-row test split:

- **SHAP** (`KernelExplainer`) over the 34 engineered features, with [CLS] held fixed at the per-sample value (200 background samples; 991 explained).

TABLE III

STAGE 2 THREE-PHASE UNFREEZING SCHEDULE.

Phase	Trainable	Epochs	LR
1 – Head only	MLP (224k)	5	1×10^{-3}
2 – Top-2 unfrozen	MLP + 2 blks (14.4M)	8	2×10^{-5}
3 – Full model	All (66.6M)	5	5×10^{-6}

TABLE IV

STAGE 1 V3 PER-CLASS RESULTS ON THE COMBINED TEST SET ($n=3,531$). MACRO-F1 = 0.763, ACCURACY = 78.7%.

Class	Precision	Recall	F1	Support
Claim	0.83	0.78	0.80	1,739
Evidence	0.45	0.59	0.51	395
Opinion	0.88	0.89	0.88	244
Background	0.86	0.84	0.85	1,153
Macro avg	0.75	0.78	0.76	3,531
Weighted avg	0.80	0.79	0.79	3,531

- **Permutation importance** (AUC drop when each feature is shuffled; 5,000-row sample for tractability).
 - **Mutual information** (`sklearn`, model-free).
- Per-method ranks are aggregated with *Borda count*: the rank-mean across methods. Lower mean rank = more consistently important across attribution paradigms. Token-level explanations use Integrated Gradients [11] on the embedding layer (via `captum`) and LIME [12] as a perturbation baseline.

V. EXPERIMENTAL RESULTS

A. Stage 1 — Epistemic Classification

Stage 1 reaches a best validation macro-F1 of **0.766** after 5 epochs (early stopping triggered at epoch 7). On the held-out combined test set (3,531 sentences), it scores macro-F1 **0.763** and accuracy **78.7%**. Per-class results are shown in Table IV; Evidence remains the weakest class (F1=0.51) but is dramatically improved over the unweighted baseline thanks to focal loss and oversampling.

a) Out-of-distribution generalisation: On the IBM-only held-out test (878 sentences containing only Claim and Evidence labels), Stage 1 scores macro-F1 **0.262** (Claim F1 0.51, Evidence F1 0.54). The macro figure is depressed by unavoidable zeros on Opinion and Background (no support); the binary Claim-vs-Evidence comparison is closer to F1 0.52, indicating that relative rankings transfer but absolute calibration does not. We discuss this in §VI.

B. Stage 2 — Citation-Need Scoring and Ablation

We train two Late-Fusion variants on identical splits: (a) **Full v3** (29 engineered + 5 epistemic, 34-dim auxiliary features), and (b) **No-epi** (29 engineered only). Both use the same backbone and three-phase schedule. Final test results on the official 991-sentence WikiSQE test split are shown in Table V.

The no-epi variant outperforms the full pipeline on every metric, with the ROC-AUC gap of -0.010 comfortably outside

TABLE V

STAGE 2 RESULTS ON THE WIKISQE TEST SPLIT (991 SENTENCES).
NEGATIVE ABLATION: REMOVING THE STAGE 1 EPISTEMIC SIGNAL
 SLIGHTLY IMPROVES ALL METRICS. BRIER SCORE: LOWER IS BETTER.

Variant	AUC	F1	Acc	AP	Brier
Full v3 (34-dim aux)	0.7270	0.6548	0.6660	0.7045	0.2353
No-epi (29-dim aux)	0.7365	0.6617	0.6781	0.7144	0.2133
Δ (full – no-epi)	-.0095	-.0069	-.0121	-.0099	+.0220

TABLE VI

TOP-12 FEATURES BY BORDA RANK ACROSS SHAP, PERMUTATION IMPORTANCE, AND MUTUAL INFORMATION ($n=991$ TEST SENTENCES).
 FAMILY ABBREVIATIONS: *Base* = BASELINE 12-FEATURE SET; *Synt* = V2 SYNTACTIC; *Verb* = V2 VERB CLASSES; *Num* = V2 NUMERICAL; *Ent* = V2 ENTITY RATES; *Epi* = STAGE 1 EPISTEMIC.

#	Feature	Family	SHAP	Perm	MI	Borda
1	sentence_length	Base	0.0256	0.0072	0.0163	2.33
2	n_clauses	Synt	0.0113	0.0001	0.0053	9.33
3	rate_causal_markers	Verb	0.0110	0.0000	0.0261	9.67
4	n_proper_nouns	Base	0.0109	0.0011	0.0005	11.33
5	rate_dollar	Num	0.0105	0.0000	0.0267	11.33
6	dep_tree_depth	Synt	0.0101	0.0007	0.0041	13.67
7	n_subjective_pron	Base	0.0103	0.0001	0.0074	14.00
8	n_hedge_verbs	Base	0.0109	0.0001	0.0000	14.00
9	epi_confidence	Epi	0.0111	-0.0000	0.0079	14.33
10	n_hedge_adv	Base	0.0102	0.0002	0.0061	14.33
11	epi_prob_Background	Epi	0.0113	0.0002	0.0000	15.00
12	n_passive	Base	0.0118	-0.0000	0.0000	16.33
16	epi_prob_Opinion	Epi	0.0107	-0.0003	0.0152	17.33
30	epi_prob_Claim	Epi	0.0097	0.0006	0.0000	21.67
33	epi_prob_Evidence	Epi	0.0103	-0.0004	0.0000	28.33

noise. This is the opposite of our pre-registered hypothesis **H3**; we discuss the interpretation in §VI.

C. Triangulated Feature Importance (RQ2)

Table VI reports the top features by Borda-rank aggregation across SHAP, permutation importance, and mutual information. `sentence_length` is the unanimous top feature (Borda 2.33), followed by `n_clauses`, `rate_causal_markers`, `n_proper_nouns`, and `rate_dollar` — a mix of the length, syntactic, and numerical-content families.

a) *The H1 verdict: epistemic features lose to surface features.*: Of the five epistemic signals only `epi_confidence` cracks the top 10. `epi_prob_Claim` and `epi_prob_Evidence` — the two probabilities our hypothesis specifically predicted to dominate — end up at ranks 30 and 33 respectively (out of 34). Hypothesis **H1** is therefore *rejected*: in this dataset and architecture, `sentence_length` alone outranks every Stage 1 output.

D. Class-Conditional SHAP

Aggregated rankings can hide class-specific dynamics. We therefore compute SHAP separately for sentences predicted to belong to each epistemic class (Table VII). The story changes substantially: *Claim*-typed sentences are driven by `n_numbers`, `rate_ent_total`,

TABLE VII

TOP-3 FEATURES BY MEAN |SHAP| WITHIN EACH PREDICTED EPISTEMIC CLASS. DIFFERENT CLASSES ARE DRIVEN BY DIFFERENT FEATURE FAMILIES.

Class	Top 1	Top 2	Top 3
Claim ($n=70$)	<code>n_numbers</code>	<code>rate_ent_total</code>	<code>rate_ent_pers.</code>
Evidence ($n=289$)	<code>sent_length</code>	<code>n_hedge_verbs</code>	<code>rate_ent_org</code>
Background ($n=617$)	<code>sent_length</code>	<code>rate_ent_date</code>	<code>n_hedge_adv</code>

TABLE VIII

THREE-AXIS COMPARISON: CLASSIFICATION, LATENCY, AND INTERPRETABILITY. LATENCY REPORTED PER SENTENCE ON THE SAME RTX 2080 Ti.

Model	Params	AUC	F1 / Lat.	Interpretability
EpiCite (ours)	66.5M	.727	.655 / 0.78 ms	✓ SHAP, IG, LIME, class-cond., calibration
Qwen-0.5B-Instruct	494M	.539	.239 / 8.43 ms	× Black-box; CoT post-hoc, not faithful

and `rate_ent_person` (numerical and named-entity content); *Evidence*-typed sentences by `sentence_length`, `n_hedge_verbs`, and `rate_ent_org`; *Background*-typed sentences by `length`, `rate_ent_date`, and hedge adverbs. The Opinion class has only 15 test rows and is omitted.

E. Calibration

Reliability diagrams indicate that the Full v3 variant is poorly calibrated ($ECE = 0.336$); the No-epi variant is markedly better ($ECE = 0.267$). Both over-confidently predict the positive class; we recommend temperature scaling [13] as a follow-up.

F. Stratified Error Analysis

Of 991 test sentences, EpiCite errs on 331 (33.4%). Error rate is highest for predicted-Opinion sentences (53.3%; only 15 cases), and short sentences (< 12 tokens, 36.2%). Confidence is well calibrated as a *relative* signal: errors fall from 44.8% in the 0.5–0.7 confidence band to 16.8% above 0.95.

a) *Cascade analysis*: Of all Stage 2 errors, only 6.3% (21 cases) coincide with low Stage 1 confidence (< 0.50); the remaining 93.7% (310 cases) occur when Stage 1 was confident. Improving Stage 1 calibration would therefore at best fix $\sim 6\%$ of Stage 2 mistakes — consistent with the small ablation Δ in Table V.

G. Comparison with Qwen-0.5B-Instruct (RQ3)

We benchmark against Qwen-0.5B-Instruct (494M parameters; $3.7\times$ larger than our pipeline’s 133.5M including Stage 1 + Stage 2) under three prompt strategies on a 3,000-sentence subsample of the test set. The best Qwen prompt (zero-shot structured) reaches AUC 0.539 / F1 0.239. EpiCite reaches AUC 0.727 / F1 0.655 with $\sim 11\times$ lower latency (Table VIII).

a) *Error overlap*: On 991 shared test sentences: both correct 35.6%, both wrong 15.3%, EpiCite-only-wrong 18.1%, Qwen-only-wrong 31.0%. Cohen’s kappa on the error sets is -0.005 , indicating near-independent failure modes — i.e., a low-cost ensemble could plausibly recover 70% combined coverage.

H. Use Cases

a) *Wikipedia article audit*: A demo article with seven sentences (two ground-truth flagged) yields 57% sentence-level agreement with the ground truth, with two false-positive flags on opinion-laden but non-citable lines (e.g. “Many visitors find it beautiful at night.”). The audit thereby surfaces *exactly* the kind of subjective sentence epistemic typing was intended to catch.

b) *MCP server*: We expose the pipeline through three Model Context Protocol endpoints: `analyze_sentence` (returns epistemic class, probabilities, citation score), `analyze_paragraph` (per-sentence batch + aggregate), and `explain_prediction` (returns IG token weights). Sample latency for a 4-sentence paragraph: ~ 15 ms end-to-end.

VI. DISCUSSION

a) *Why does the no-epi variant beat the full pipeline?*: Three plausible mechanisms. **(i) Information redundancy**. The Stage 1 logits are computed by a DistilBERT encoder; the same encoder also produces [CLS] for Stage 2. Consequently the epistemic signal is almost certainly encoded in [CLS] already, and the explicit five-dim augmentation adds noise without information. **(ii) Calibration drag**. Stage 1’s IBM-ODD calibration breakdown (§5.A) suggests that the absolute probability values, while locally meaningful for the Stage 1 task, are noisy regressors for the Stage 2 task. **(iii) Domain gap**. Stage 1 is trained on argumentative IBM/UKP sentences whose feature distributions differ markedly from encyclopedic WikiSQE; the explicit handoff transmits this distribution shift directly into the scorer.

b) *Does this falsify the design?*: No — but it changes the story. The two-stage architecture’s *accuracy* contribution is essentially zero. Its *interpretability* contribution remains substantial: class-conditional SHAP analyses (Table VII), the audit use case, and the MCP `explain_prediction` endpoint all depend on having the epistemic label as a first-class output. We therefore now position EpiCite as an *interpretability-first* pipeline, with the no-epi variant available as the strict accuracy baseline.

c) *Why does sentence_length dominate?*: WikiSQE positives are skewed long — most are claim-like sentences embedding factual material (numbers, dates, named entities), whereas negatives are often short Wikipedia connectives, headings, or list fragments. `sentence_length` silently encodes much of this variance, which is precisely the *kind* of confound that motivates the multi-track explainability suite: had we relied only on SHAP we would have over-credited length; mutual information surfaces the genuinely informative numerical features (`rate_dollar`, `rate_causal_markers`,

`rate_universal_quant`) that correlate with claims regardless of length.

d) *Hypothesis verdict.*: **H1 (epistemic dominates surface)**: *Rejected*. Sentence length is the unanimous top feature; epistemic probabilities cluster at ranks 9–33. **H2 (better interpretability than end-to-end)**: *Supported*. We provide three feature-level views (SHAP, permutation, MI), one token-level view (IG), and one class-conditional view, in addition to the audit and MCP endpoints — none of which is straightforward in an end-to-end binary classifier. **H3 (Stage 1 signal helps)**: *Rejected*. Removing it *improves* every metric on the test split.

e) *Limitations.*: (i) *Domain*. All quantitative results live on Wikipedia-style text; cross-domain validation on news / student essays is left to future work. (ii) *Annotation noise*. A 40-sentence pilot of borderline disagreements suggests non-trivial label noise in WikiSQE; the noise rate has not yet been hand-coded. (iii) *Calibration*. Both variants are over-confident ($ECE \geq 0.27$); temperature scaling should be applied before deployment. (iv) *Evidence class*. Despite focal loss and IBM augmentation, Evidence remains the weakest Stage 1 class (F1 0.51) due to lexical heterogeneity and the absence of canonical surface markers.

VII. CONCLUSION

We presented EpiCite, a two-stage pipeline that classifies sentence epistemic type and predicts citation need on Wikipedia text. A focal-loss DistilBERT trained on a class-balanced IBM/UKP merge reaches macro-F1 0.763 across four epistemic classes. A Late-Fusion citation scorer over [CLS] plus 29 engineered features reaches ROC-AUC 0.727 / F1 0.655 on WikiSQE, outperforming a $3.7\times$ -larger Qwen-0.5B-Instruct LLM by 0.19 AUC at $\sim 11\times$ lower latency, and with near-independent error modes ($\kappa = -0.005$).

A triangulated feature analysis (SHAP + permutation + mutual information) together with a no-epi ablation refutes two of our three pre-registered hypotheses: `sentence_length` dominates every epistemic feature, and removing the Stage 1 signal slightly *improves* the scorer. We take this as evidence that the explicit two-stage design earns its keep through *interpretability* (class-conditional SHAP, IG explanations, the MCP audit interface) rather than through accuracy. Future work focuses on cross-domain transfer to news and student essays, calibration via temperature scaling, ensembling with Qwen given the favourable error overlap, and human noise estimation on WikiSQE labels.

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